Type 2 diabetes

What is type 2 diabetes?

Type 2 diabetes is a condition in which the body becomes resistant to the normal effects of insulin and gradually loses the capacity to produce enough insulin in the pancreas. The condition has strong genetic and family-related (non-modifiable) risk factors and is also often associated with modifiable lifestyle risk factors. We do not know the exact genetic causes of type 2 diabetes. People may be able to significantly slow or even stop the progression of the condition through changes to diet and increasing the amount of physical activity they do.

Type 2 diabetes:

- Is diagnosed when blood glucose levels are high due to insulin produced by the pancreas not working effectively and/or the cells of the body do not respond to insulin effectively (known as insulin resistance), over time the condition progresses and the pancreas does not produce enough insulin (reduced insulin production)
- Represents 85–90 percent of all cases of diabetes
- Usually develops in adults over the age of 45 years but is increasingly occurring in younger age groups including children, adolescents, and young adults
- Is more likely in people with a family history of type 2 diabetes or from particular ethnic backgrounds
- Can develop from the long-term use of corticosteroids to treat other health conditions
- For some, the first sign may be a complication of diabetes such as a heart attack, vision problems or a wound that does not heal well

- Is managed with a combination of regular physical activity, healthy eating, and weight reduction. As type 2 diabetes can be progressive, many people will need oral medications and/or insulin injections in addition to lifestyle changes over time

What happens with type 2 diabetes?

Type 2 diabetes develops over a long period of time (years). During this period of time insulin resistance starts, this is where the insulin is increasingly ineffective at managing the blood glucose levels. As a result of this insulin resistance, the pancreas responds by producing greater and greater amounts of insulin, to try and achieve some degree of management of the blood glucose levels[i].

As insulin overproduction occurs over a very long period of time, the insulin-producing cells in the pancreas wear themselves out, so that by the time someone is diagnosed with type 2 diabetes, they have lost 50 – 70% of their insulin-producing cells. This means type 2 diabetes is a combination of ineffective insulin and not enough insulin. Lifestyle changes may be able to slow this process in some people.

Initially, type 2 diabetes can often be managed with healthy eating and regular physical activity. Over time many people with type 2 diabetes will also need tablets, and/or non-insulin injectable medications and many eventually require insulin injections. It is important to note that this is normal, and taking tablets, non-insulin injectable medications or insulin as soon as they are required can result in fewer long-term complications.

What causes type 2 diabetes?

Diabetes runs in the family. If you have a family member with diabetes, you have a genetic disposition to the condition.

While people may have a strong genetic disposition towards type 2 diabetes, the risk is greatly increased if people display a number of modifiable lifestyle factors including high blood pressure, being overweight, not being active enough, eating a poor diet and having the classic 'apple shape' body where extra weight is carried around the waist.

People are at a higher risk of getting type 2 diabetes if they:

- Have a family history of diabetes
- Are older (over 55 years of age) – the risk increases as we age
- Are over 45 years of age and are overweight
- Are over 45 years of age and have high blood pressure
- Are over 35 years of age and are from an Aboriginal or Torres Strait Islander background
- Are over 35 years of age and are from Pacific Island, Indian subcontinent or Chinese cultural background
- Are a woman who has given birth to a child over 4.5 kgs (9 lbs), or had gestational diabetes when pregnant, or has a condition known as Polycystic Ovarian Syndrome (PCOS)
- Have mother who had gestational diabetes when they were in utero
- Have been prescribed long-term use of a steroid for treating some cancers, including leukemia, or conditions that cause inflammation such as severe asthma, cystic fibrosis, arthritis, and inflammatory bowel disease.

Check your risk – answer 11 short questions to better understand your risk of developing type 2 diabetes[ii].

Read our section on preventing type 2 diabetes.

Symptoms of type 2 diabetes

Many people with type 2 diabetes have no symptoms. As type 2 diabetes is commonly (but not always) diagnosed at a later age, sometimes signs are dismissed as a part of 'getting older'. In some cases, by the time type 2 diabetes is diagnosed, the long-term complications of diabetes may already be present.

Symptoms include:

- Being excessively thirsty
- Passing more urine
- Feeling tired and lethargic
- Always feeling hungry
- Having cuts that heal slowly
- Itching, skin infections
- Blurred vision
- Gradually putting on weight, or some people will lose weight
- Mood swings

- Headaches
- Feeling dizzy
- Leg cramps

Managing type 2 diabetes

While there is currently no cure for type 2 diabetes, the condition can be managed through lifestyle modifications and medication. Effectively managing diabetes is the best way to prevent diabetes-related complications.

If you have recently been diagnosed with type 2 diabetes or have a family member with type 2 diabetes, view information here on managing type 2 diabetes.

Diabetes remission

Research shows that it is possible for some people with type 2 diabetes to achieve type 2 diabetes remission[iii].

We define type 2 diabetes remission as an HbA1c of under 6.5% (48mmol/mol) for at least three months without the need for glucose-lowering medications.

The most common ways people have achieved remission is by achieving substantial weight loss following very intensive dietary changes or through bariatric surgery.

Remission is not achievable for everyone with type 2 diabetes. In all studies of intensive dietary modifications, less than half of participants were able to achieve remission after one year.

Remission does not mean type 2 diabetes is cured or reversed – it simply means that people have blood glucose levels below the type 2 diabetes levels. It is important that people in remission continue to access regular diabetes monitoring at least annually and keep up their Annual Cycle of Care health care checks. Ongoing monitoring is required as some of the macrovascular and microvascular damage which cause long term complications has already begun.

People with type 2 diabetes who want to attempt diabetes remission need to do so in close consultation with their diabetes healthcare team, as intensive dietary and weight changes need careful management, monitoring and support.

People who do not achieve or sustain remission should not feel that they have 'failed'. The health benefits of weight loss and a reduction in HbA1c are significant even if remission does not occur, as these reduce the risk of developing diabetes-related complications and may lead to reducing or stopping glucose-lowering medications.

Diabetes Australia has developed a position statement on diabetes remission[iv] to help people with diabetes and health professionals make informed choices.

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[i] Leszek Szablewski. (2011). *Glucose homeostasis and insulin resistance*. Bentham Science. https://benthambooks.com/book/9781608051892/ p126-128

[ii] https://www.health.gov.au/resources/apps-and-tools/the-australian-type-2-diabetes-risk-assessment-tool-ausdrisk

[iii] Lean, M. E., Leslie, W. S., Barnes, A. C., Brosnahan, N., Thom, G., McCombie, L., Peters, C., Zhyzhneuskaya, S., Al-Mrabeh, A., Hollingsworth, K. G., Rodrigues, A. M., Rehackova, L., Adamson, A. J., Sniehotta, F. F., Mathers, J. C., Ross, H. M., McIlvenna, Y., Stefanetti, R., Trenell, M., & Welsh, P. (2018). Primary care-led Weight Management for Remission of Type 2 Diabetes (DiRECT): an open-label, cluster-randomised Trial. *The Lancet*, *391*(10120), 541–551. https://doi.org/10.1016/s0140-6736(17)33102-1

[iv] *StackPath*. (n.d.). www.diabetesaustralia.com.au. https://www.diabetesaustralia.com.au/wp-content/uploads/2021_Diabetes-Australia-Position-Statement_Type-2-diabetes-remission_2.pdf